

PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

**Anonymous Channels for Workplace Communication: Design, Implementation,
and Survey Analysis of Chatify**

submitted in partial fulfilment of the requirements for the award of degree of

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled "Anonymous Channels for Workplace Communication: Design, Implementation, and Survey Analysis of Chatify" in partial fulfilment of the requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the Department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of our own work carried out under the supervision of Dr. Ashish. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidates is correct to the best of my knowledge and belief.

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**Paras Sanserwal
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ABSTRACT

Although workplace collaboration applications such as Slack and Microsoft Teams have transformed organizational communication, employees remain reluctant to participate in company-wide or team channels due to fear of judgement and social evaluation. This project presents Chatify, a web-based anonymous chat application developed using React.js, Node.js/Express, Socket.io, and MongoDB.

A two-week pilot study was conducted with 60 invited users, of whom 52 responded (87% response rate). Survey results showed that 88% of participants felt more comfortable communicating anonymously, 83% reported asking questions they would not have raised otherwise, and 81% would recommend Chatify to a colleague. The platform achieved a usability score of 86% and a trust-in-anonymity rating of 74%.

The study also identified key challenges including misuse of anonymous input, lack of accountability, toxic behavior, and spam. Proposed mitigations include a hybrid anonymity model with encrypted identity audit logs, a two-tiered content moderation pipeline, keyword filtering, rate limiting, session-scoped pseudonym rotation, and a Verified Expert badge system. This report documents the design, implementation, findings, and roadmap for Chatify.

Keywords:

Anonymous communication, workplace chat, psychological safety, Chatify, organizational communication, real-time messaging, moderation, misuse prevention

1. Introduction

1.1 Background

Digital workplace communication tools such as Slack and Microsoft Teams are now ubiquitous in organizational infrastructure, enabling both organization-wide and team-specific channels. Despite this adoption, the behavioral trend of not speaking up in company-wide or team communication channels is still widely reported. Factors inhibiting workplace communication that have been documented in the organizational behavior literature include social anxiety, fear of negative evaluation, impostor syndrome, and perceived power asymmetry between junior contributors and senior leadership.

Edmondson's pioneering work on psychological safety demonstrates that team performance improves when people have the freedom to take interpersonal risks — reporting problems, asking questions, reporting mistakes, giving and receiving feedback, and proposing new ideas. Anonymous channels are one mechanism to increase psychological safety because they decouple people's identity from their contributions: a message is evaluated only for its content and not for its sender's identity.

1.2 Problem Statement

Despite the theoretical benefits of anonymity, anonymous platforms introduce well-documented risks including lack of accountability, which can lead to harassment, misinformation, and toxic speech — as demonstrated by platforms such as Yik Yak and early versions of Blind. The central design challenge is therefore to preserve the psychological safety benefits of anonymity while providing sufficient organizational guardrails to prevent misuse.

1.3 Objectives

- To design and implement Chatify, a web-based anonymous workplace messaging application.
- To conduct a structured pilot study measuring the effect of anonymous communication on participation behavior.
- To identify key challenges posed by anonymous communication in organizational contexts.
- To propose concrete mitigations for each identified challenge.

1.4 Scope

The project covers the full-stack development of Chatify using React.js, Node.js, Socket.io, and MongoDB, followed by a two-week organizational pilot with 52 participants. The scope includes quantitative Likert-scale analysis and qualitative thematic coding of open-ended survey responses, as well as a roadmap for future enhancements.

2. Methodology

2.1 Study Design

The tool was piloted within a medium-sized technology company over a two-week period. At the end of the test period, users completed a structured survey instrument designed to capture behavioral change, psychological safety perceptions, usability assessment, and misuse concerns.

The survey comprised ten Likert-scale items (1 = Strongly Disagree, 5 = Strongly Agree) and two open-ended questions. The ten items were categorized across four dimensions: communication behavior change, psychological safety, usability, and misuse concerns.

2.2 Participant Demographics

Invitations were sent to 60 employees, of whom 52 complete and valid responses were received, yielding an overall response rate of 87%. Participant roles included software engineers (45%), product managers (21%), QA analysts (20%), and HR employees (14%). Ages ranged from 22 to 45. 61% of respondents identified as male, 36% as female, and 3% did not specify.

2.3 Survey Instrument

Survey items were modeled after established scales in organizational communication and human-computer interaction (HCI) research. Two academic advisors reviewed the instrument for face validity. Open-ended responses were analyzed using thematic coding with two independent coders, achieving strong inter-rater reliability ($\kappa = 0.81$).

2.4 Data Analysis

Quantitative data were analyzed by computing mean Likert scores and agreement percentages (proportion of respondents selecting 4 or 5) for each survey item. Qualitative open-ended responses were coded into themes inductively by two independent reviewers, with disagreements resolved through discussion.

3. Tools and Technologies

3.1 Frontend — React.js

The frontend of Chatify is a single-page application (SPA) built with React.js following the component-based design pattern. React manages channel state, message threads, and user authentication flows. The component-driven architecture allows modular development and easy feature extension.

3.2 Backend — Node.js / Express

The backend is a Node.js/Express application that exposes a REST API for user registration, login, channel management, and message storage. Express provides a lightweight, unopinionated framework for routing and middleware management, making it well-suited for rapid development of RESTful endpoints.

3.3 Real-Time Communication — Socket.io

Bidirectional, low-latency messaging is achieved via the Socket.io WebSockets library. Unlike HTTP polling, Socket.io establishes a persistent connection, enabling sub-second message delivery. The library handles connection lifecycle, room-based broadcasting (mapped to channels), and reconnection logic automatically.

3.4 Database — MongoDB with Mongoose

Message and user data are persisted in MongoDB, a document-oriented NoSQL database. The Mongoose ODM (Object Document Mapper) provides schema definitions, validation, and query helpers, bridging Node.js application logic and the MongoDB storage layer.

3.5 Development & Deployment Tools

- Git — version control for collaborative development.
- npm — package management for both frontend and backend dependencies.
- Postman — API testing and documentation during development.
- VS Code — primary code editor used by the development team.

4. Implementation

4.1 System Architecture Overview

Chatify is a web application that allows authenticated users to join named group chat channels and send messages without revealing their user identity to other channel members. The primary privacy guarantee is enforced server-side: when a message is received by the Node.js/Express API server, the sender's identity metadata is stripped before the message is broadcast to the channel via the Socket.io real-time layer. This design ensures that even packet-level network analysis of client traffic cannot reveal sender identity.

The architecture comprises three primary layers: (1) the React.js client (browser), (2) the Node.js/Express API server with integrated Socket.io real-time layer, and (3) the MongoDB database. The API server acts as the trust boundary where identity is anonymized.

4.2 Key Features Implemented

- Peer-anonymous group messaging across named channels — messages are stripped of sender metadata before broadcast.
- Real-time delivery via Socket.io WebSocket events, providing sub-second message latency.
- Channel-based messaging with persistent message history stored in MongoDB.
- Username and password authentication providing organizational context while maintaining in-channel anonymity.
- Session-scoped pseudonym rotation — each user is assigned a randomly generated pseudonym (e.g., "Anonymous Falcon") for the duration of a session, enabling threaded conversation coherence while preventing stylometric identity tracking across sessions.

4.3 Anonymity Enforcement

The anonymity model is peer-anonymous: administrators who manage the platform retain the ability (through a server-side audit log) to associate messages with senders in formal escalation scenarios, while peers in the channel see only pseudonymous labels. This hybrid approach balances psychological safety with organizational accountability.

4.4 Limitations of the Current Prototype

The current prototype does not yet implement a content moderation panel, automated keyword filtering, a peer reporting workflow, or rate limiting on message volume. These are identified as priority items for the next development cycle and are documented in Chapter 6.

5. Major Findings / Outcomes / Results

5.1 Quantitative Survey Results

The mean Likert ratings and agreement percentages for the eight core survey items are presented in Table I below.

Survey Question	Mean Score	Agree %
Q1: Comfortable without anonymity	2.3	28%
Q2: More comfortable with anonymity	4.4	88%
Q3: Asked new questions via Chatify	4.1	83%
Q4: Improves communication quality	4.0	78%
Q5: Concerned about misuse	3.2	52%
Q6: Platform easy to use	4.3	86%
Q7: Trust in identity protection	3.9	74%
Q8: Would recommend Chatify	4.2	81%

Table I: Survey Question Mean Scores (N = 52)

5.2 Key Quantitative Findings

The most striking finding is the 60 percentage-point gap between Q1 (28% of participants comfortable communicating without anonymity) and Q2 (88% comfortable with anonymity). This result strongly supports the hypothesis that anonymity substantially reduces perceived interpersonal risk of participation, consistent with prior work by Edmondson and Liang et al.

Q3 showed that 83% of participants reported asking questions they would not have asked in a non-anonymous setting — a large effect size in the organizational intervention literature. Q6 indicated 86% found Chatify easy to use (mean 4.3/5), and Q8 showed 81% would recommend it to colleagues (mean 4.2/5). The sole item falling below 50% agreement was Q1, confirming the baseline communication inhibition that the platform was designed to address.

Q5 revealed that 52% expressed concern about potential misuse, highlighting that accountability and moderation features are critical adoption enablers for the next release.

5.3 Before vs. After Comparison

Self-reported communication data showed clear improvement after the pilot. Participants who freely asked questions increased from 28% to 88%. Participants who held back from asking increased from 12% (post-pilot) versus 72% (pre-pilot). Knowledge sharing improved from 30% to 78%, and willingness to ask questions in the future increased from 35% to 81%.

5.4 Qualitative Themes

Open-ended responses yielded three major qualitative themes:

- More honest culture — "I finally felt comfortable asking about the deployment process without worrying about seeming inexperienced."
- Less social friction — "It was easier to raise concerns about unrealistic timelines when my name was not attached."
- Accountability concerns — "I worried someone might post something inappropriate without any accountability."

5.5 Challenges and Proposed Mitigations

Six challenges were identified through survey data, prior literature, and internal design review. Table II summarizes each challenge alongside its implementation status and proposed mitigation strategy.

Challenge	Status	Proposed Mitigation
Misuse / Inappropriate Content	Not yet addressed	Keyword filter + admin moderation panel
Reduced Accountability	Not yet addressed	Hybrid: encrypted identity (admin-only)
Toxic Behavior / Harassment	Not yet addressed	Peer reporting + escalation workflow
Spam / Message Flooding	Not yet addressed	Rate limiting (max 5 msg/min/user)
Long-term Identity Tracking	Implemented	Session-scoped pseudonym rotation
Lack of Authoritative Answers	Not yet addressed	Verified Expert badge on admin replies

Table II: Challenges and Proposed Mitigations

6. Conclusion and Future Scope

6.1 Conclusion

This project introduced Chatify, a web-based anonymous workplace communication platform built with React.js, Node.js/Express, Socket.io, and MongoDB. A two-week pilot study with 52 participants demonstrated strong empirical support for the hypothesis that structured anonymity increases workplace communication participation. Key outcomes included 88% of participants feeling more comfortable communicating anonymously, 83% reporting they asked questions they would otherwise have withheld, and 81% willing to recommend the platform.

The current implementation is a functional proof of concept that validates the core premise. However, it does not yet fully address the accountability, moderation, and anti-abuse needs identified by 52% of respondents. The roadmap includes a hybrid anonymity model with cryptographic audit logs, keyword filtering, rate limiting, peer reporting, session-scoped pseudonym rotation (already implemented), and a Verified Expert badge for authoritative answers.

6.2 Future Scope

- Multi-organizational study: Extend the pilot across multiple organizations of varying sizes and cultures to evaluate generalizability.
- Hybrid anonymity model: Implement cryptographically bound sender-message associations in a sealed server-side audit log accessible only through a formal escalation process.
- Content moderation pipeline: Deploy a two-tiered system comprising (1) automated keyword blacklist filtering and (2) community-driven peer reporting with a moderator escalation queue.
- Rate limiting: Enforce a maximum of 5 messages per minute per user to reduce spam flooding.
- Verified Expert badge: Allow administrators to attach verified credentials to replies to specific questions, addressing the "lack of authoritative answers" challenge.
- Longitudinal study: Conduct a longer-term study (minimum 3 months) to assess cumulative effects of anonymity on team cohesion, culture, and moderation load.

- Controlled A/B experiment: Compare full-anonymity, hybrid-anonymity, and identified conditions within a controlled experimental design.

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Appendices

Appendix A: Survey Instrument

The following ten Likert-scale items (1 = Strongly Disagree, 5 = Strongly Agree) were administered to participants at the end of the two-week pilot:

- Q1: I feel comfortable asking questions or raising concerns in company-wide channels without anonymity.
- Q2: I feel more comfortable communicating in Chatify because my identity is anonymous.
- Q3: I asked questions or shared concerns on Chatify that I would not have shared otherwise.
- Q4: Chatify improves the overall quality of communication in our team.
- Q5: I am concerned that anonymous messaging could be misused by some users.
- Q6: The Chatify platform was easy to use.
- Q7: I trust that Chatify adequately protects my identity.
- Q8: I would recommend Chatify to a colleague.

Open-ended questions:

- Describe a specific situation where anonymous communication helped or hindered your work during the pilot.
- What feature or policy change would most improve your confidence in using Chatify?

Appendix B: Technology Stack Summary

Component	Technology	Purpose
Frontend	React.js	SPA UI & state management
Backend	Node.js / Express	REST API server
Real-time Layer	Socket.io	WebSocket messaging
Database	MongoDB + Mongoose	Data persistence